

Classical Test Theory & Reliability Analysis

Method Dossier | Basic Level

1. Method Description

Classical Test Theory (CTT) decomposes observed score X into true score T and error E : $X = T + E$. Assumptions: $E[E]=0$, $r(T,E)=0$, $r(E_i,E_j)=0$. Reliability = $\text{Var}(T)/\text{Var}(X)$.

Cronbach Alpha (Internal Consistency)

$\alpha = (k/(k-1)) * (1 - \sum(\text{Var}_i)/\text{Var}_{\text{total}})$. Thresholds: >0.7 acceptable, >0.8 good, >0.9 excellent.

Test-Retest Reliability

Correlation between two administrations. Measures temporal stability. Ideally 2–4-week interval.

Split-Half Reliability

Correlates two test halves. Spearman-Brown correction: $r_{\text{total}} = 2 * r_{\text{half}} / (1 + r_{\text{half}})$.

2. Field Examples

Psychology: Depression scale (20 items), $\alpha = 0.92$. Item-total correlations 0.35-0.72.

Medicine: SF-36 health survey, test-retest $r = 0.85$, split-half $r = 0.82$, corrected $r = 0.90$.

Social Sciences: Political cynicism scale (5 items), $\alpha = 0.68$, improved to 0.76 after removing one item.

3. Assumptions

$X = T + E$, $E[E] = 0$, $r(T,E) = 0$, $r(E_i,E_j) = 0$

Minimum 2 items per scale (3-5 recommended)

Items should measure the same construct

4. Interpretation and Reporting

Report α with 95% CI, sample size, number of items

Item statistics: mean, SD, item-total correlation, α -if-deleted

For test-retest: report time interval

5. Code Examples

R:

```
library(psych)
alpha(items, check.keys=TRUE)
splitHalf(items)
```

Python:

```
from pingouin import cronbach_alpha
alpha, ci = cronbach_alpha(data)
```

6. Extension

McDonald Omega

More robust than alpha when tau-equivalence is violated. Based on factor loadings.

Interrater Reliability

Cohen Kappa (2 raters, nominal), ICC (metric), Fleiss Kappa (>2 raters).

7. Exercises

Exercise 1: Anxiety scale (10 items, $n=100$), $\alpha = 0.72$. Item 4 alpha-if-deleted = 0.78. Interpret. Remove item?

Exercise 2: Motor performance test-retest: $r=0.95$ (1 day), $r=0.68$ (4 weeks). Interpret difference. True reliability?

Exercise 3: 20-item test, split-half $r=0.70$. Compute Spearman-Brown corrected reliability.